

Sahana Aster KŌWHAI, Ph.D.

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Associate Professor of Palaeoclimatology, Te Tari Tātai Arowhenua — Department of Geology

University of Otago — Ōtākou Whakaihu Waka,

Antarctic Research Centre, Te Herenga Waka, Ōtepoti Dunedin, Aotearoa New Zealand

SHORT BIO

Sahana Aster Kōwhai is a **fictional** palaeoclimatologist whose *research* reconstructs Southern Ocean change from marine sediment archives.

RESEARCH FOCUS

Southern Ocean palaeoceanography across glacial terminations, with an emphasis on multiproxy reconstructions, reproducible chronology and equitable Antarctic field partnerships.

ACADEMIC APPOINTMENTS / EXPERIENCE

- Associate Professor of Palaeoclimatology**
University of Otago — Ōtākou Whakaihu Waka, Department of Geology
Feb 2021 – Present
Ōtepoti Dunedin, Aotearoa New Zealand
 - Cross-appointed affiliate of the Antarctic Research Centre, Te Herenga Waka.
- Senior Lecturer in Marine Geology**
University of Tasmania, Institute for Marine and Antarctic Studies
Jul 2016 – Jan 2021
nipaluna Hobart, Australia

EDUCATION

- Ph.D. in Marine Geology**
Australian National University
2010 – 2014
Canberra, Australia
 - Thesis: Foraminiferal Mg/Ca and Southern Ocean warming across Termination I.
- M.Sc. in Earth Sciences (First Class Honours)**
University of Cape Town
2008 – 2009
Cape Town, South Africa
- B.Sc. in Geology and Oceanography**
University of Madras
2005 – 2008
Chennai, India

TEACHING

- Course coordinator and lecturer**
University of Otago — Ōtākou Whakaihu Waka
2021 – Present
Ōtepoti Dunedin, Aotearoa New Zealand

Course	Programme	Topics	Points
GEOL 273	B.Sc. Geology	Palaeoclimate archives; proxy calibration	18
GEOL 472	B.Sc. (Hons)	Marine cores; age-depth modelling	18
OCEN 405	M.Sc. Oceanography	Southern Ocean circulation; uncertainty	20

EDITORIAL, REVIEWING & CONFERENCE SERVICE

- Associate Editor**
Paleoceanography and Paleoclimatology, Marine palaeoclimate [IF: 3.4, Q1]
2023 – Present
- Scientific Steering Committee**
Southern Ocean Observing System, 2022–2026
- Panel reviewer**
Royal Society Te Apārangi
2021 – Present
- Reviewer**
Quaternary Science Reviews

RESEARCH PROJECTS

- TIDE — Termination Ice Dynamics and Ecosystems**
Royal Society Te Apārangi, Principal investigator
2024 – 2027
 - Multiproxy reconstruction of Southern Ocean fronts across Termination II, with voyage support from the [Southern Archive Voyage Consortium](#) and sampling decisions recorded in a [field note](#).
- Open Core Chronologies**
Australian Antarctic Program Partnership, Co-investigator
2022 – 2025
 - Reusable age-depth models and provenance for legacy sediment cores.

Awards & Scholarships

- Early Career Research Excellence Award

Geoscience Society of New Zealand

2019
- SCAR Visiting Fellowship

Scientific Committee on Antarctic Research

2017
Cambridge, United Kingdom

Student Supervision (Theses & Internships)

Current supervision: projects span marine geochemistry, chronology and open polar data.

Level	#	Notes (scope and international experience)
Doctoral candidates	4	Primary supervisor for two; co-supervisor for two
M.Sc. and honours researchers	9	Completed and current research dissertations

Additional Information

Languages: Tamil (Native), English (Fluent), te reo Māori (Intermediate learner)

Public Engagement

- Sediment stories: reading climate from ocean cores

Kōwhai Coast Learning Collective, Community workshop for secondary-school science teachers

Sep 2025
Ōamaru, Aotearoa New Zealand
- Core archive open day

Aotearoa Marine Memory Trust, Public demonstration of core scanning and age-model uncertainty

Mar 2024
Porirua, Aotearoa New Zealand
- Antarctic archives teacher residency

Polar Evidence Studio, Short course on proxy evidence for secondary-school teachers

Jul 2023
Ōtepoti Dunedin, Aotearoa New Zealand

Talks & Presentations

I=Invited, C=Conference

Invited and keynote talks

- [I1] S. A. Kōwhai. “What an open sediment record should remember”. International Conference on Paleocyanography (Bergen, Norway). Invited keynote. June 18, 2026.

Conference presentations

- [C1] S. A. Kōwhai and M. Te Rangi. “First results from the Pukaki Rise coring transect”. New Zealand Antarctic Science Conference (Te Whanganui-a-Tara Wellington, Aotearoa New Zealand). Oral presentation. Aug. 21, 2025.
- [C2] S. A. Kōwhai. “Age-depth ensembles without hidden decisions”. Australasian Quaternary Association Biennial Meeting (nipaluna Hobart, Australia). Poster presentation. Dec. 4, 2024. URL: <https://talks.example.edu/kowhai2024ensembles>.
- [C3] S. A. Kōwhai. “From core description to an auditable climate record”. Southern Archives Methods Forum (Ōtepoti Dunedin, Aotearoa New Zealand). Sept. 14, 2023.

Publications

J=Journal, C=Conference, B=Books, D=Data, R=Under Review

Journal articles (peer-reviewed)

- [J1] S. A. Kōwhai, M. Te Rangi, and C. Okafor. “Southern Ocean front migration recorded across two glacial terminations”. In: *Palaeogeography, Palaeoclimatology, Palaeoecology* 612 (2026), p. 111001.
- [J2] S. A. Kōwhai, H. Williams, and T. Ndlovu. “Transparent age-depth ensembles for Southern Ocean sediment archives”. In: *Climate of the Past* 20.8 (2024), pp. 1701–1724. URL: <https://publications.example.edu/kowhai2024chronology>.

Conference papers (peer-reviewed)

- [C1] S. A. Kōwhai and M. Te Rangi. “Automated tephra isochron matching in Southern Ocean sediment cores”. In: *Proceedings of the International Palaeocyanography Symposium*. Bergen, Norway, 2023, pp. 145–152. DOI: [10.5555/palaeo.2023.0145](https://doi.org/10.5555/palaeo.2023.0145).

Books and chapters

- [B1] S. A. Kōwhai. “Proxy agreement and disagreement in high-latitude marine archives”. In: *Southern Ocean Palaeocyanography: Methods and Records*. Ed. by L. Navarro and K. Adeyemi. Cape Town: Aster Academic Press, 2023, pp. 81–108.
- [B2] S. A. Kōwhai and C. Okafor. “Southern Ocean Foraminiferal Assemblages as Thermal Proxies”. In: *Advances in Marine Palaeoecology*. Ed. by L. Navarro. Aster Academic Press, 2022. Chap. 4, pp. 89–112.
- [B3] S. A. Kōwhai and M. Te Rangi. *Field Notes for Reciprocal Polar Science*. Ōtepoti Dunedin: Koru Open Books, 2021.

Data archives

- [D1] S. A. Kōwhai and C. Okafor. *Pukaki Rise multiproxy core chronology, version 1.0*. Aotearoa Polar Data Commons, 2025. URL: <https://data.example.edu/pukaki-rise-v1>.

Manuscripts under review

- [R1] S. A. Kōwhai and M. Te Rangi. "Shelf-water change at the Pacific Antarctic margin". In: *Quaternary Science Reviews* (2028).
- [R2] S. A. Kōwhai and T. Ndlovu. "Sea-ice retreat and productivity at the East Antarctic margin". In: *The Cryosphere* (2027).

Example record. This curriculum vitae describes a fictional scholar and is supplied only to demonstrate the template.